

Probabilistic Analysis and Quick Sort Notes

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Introduction

This technical note summarizes the tools and logic used in probabilistic analysis.

Intuition

Approach

Complexity

- Time complexity:
- Space complexity:

Code

```
class Solution {
    public int[] twoSum(int[] nums, int target) {
        HashMap<Integer, Integer> map = new HashMap<>();

        for (int i = 0; i < nums.length; i++) {
            int need = target - nums[i];

            if (map.containsKey(need)) {
                return new int[]{map.get(need), i};
            }

            map.put(nums[i], i);
        }

        return new int[]{};
    }
}
```

Reference List

1. ["CSE2331_Types_of_Probabilistic_Problems"](#) Created by Professor Painter.
2. ["Quick Sort Worksheet AU26"](#) Created by Professor Painter.

3. `"CSE2331_QuickSort_and_Probabilistic_Analysis_Homework"` Created by Professor Painter.